

What AppFlow actually is (in one sentence)

It's a **managed data-moving service** that copies data between SaaS apps (Salesforce, Slack, Zendesk, Marketo, etc.) and AWS services (S3, Redshift, Customer Profiles) — **without you writing integration code**: Amazon AppFlow is a fully-managed integration service that enables you to securely exchange data between software as a service (SaaS) applications, such as Salesforce, and AWS services, such as Amazon Simple Storage Service (Amazon S3) and Amazon Redshift.

Think of it like a plumber who already has pipes fitted to every brand of sink and every brand of drain — you just tell them "connect this sink to that drain" and they handle the fittings (authentication, API calls, pagination, retries).

The 3 building blocks you need to understand

1. Connector

A pre-built adapter for a specific app (Salesforce, Zendesk, S3, etc.) that already knows how to talk to that app's API.

2. Connection

This is the *authenticated link* to a specific account of that app. Connections store the configuration details and credentials necessary to run flows without the need to repeatedly enter information. You set this up once (e.g., log into your Salesforce dummy-data org and click "Allow"), and after that, every flow can reuse it — you don't log in again per flow.

3. Flow

This is the actual job definition — "move data from X to Y, mapped this way, on this trigger." A flow uses a connection to transfer data between a source and a destination. When you run a flow, Amazon AppFlow verifies that the data is available in the source, processes the data according to the flow configuration, and transfers the processed data to the destination.

Every flow needs 4 things configured:

- **Source** (e.g., Salesforce → Contact object)

- **Destination** (e.g., S3, or in our case Amazon Connect Customer Profiles)
- **Field mapping** (which Salesforce field → which destination field)
- **Trigger** (how it fires — see below)

The 3 ways a flow can run

- **On demand** — you click "Run flow" manually in the console.
- **Scheduled** — runs on a timer (hourly, daily, etc.) to keep data in sync.
- **Event-triggered** — runs automatically the instant something changes in the source. Your event-triggered flow runs when a change occurs to a record that you've set up to generate change event notifications — for Salesforce this is powered by Change Data Capture / Platform Events. So if a Salesforce record is edited, AppFlow can react within about a minute.

For your case (dummy data that isn't changing every second), **on-demand** is the simplest starting point — you control exactly when it pulls from Salesforce.

Why it's genuinely useful, not just "another way to call an API"

- **No code** — Create data flows to transfer data between a source and destination in minutes, without writing any code.
- **Data stays in sync automatically** if you schedule/event-trigger it.
- **Aggregation** — Bring your data together — Aggregate data from multiple sources, so you're not stuck with just one CRM feeding one destination.
- **Security by default** — everything is encrypted at rest and in transit, and it can even route traffic privately over AWS's own network instead of the public internet, via AWS PrivateLink integration — meaning your Salesforce data never has to touch the open internet on its way to AWS.
- **Auditability** — every action is logged via AWS CloudTrail integration, so you (or your mentors) can always answer "who ran this flow, and when, and what did it touch."

How permissions work (the boring but important part)

Before you can do anything, your IAM user/role needs the `AmazonAppFlowFullAccess` managed policy, **plus** an S3-specific policy if S3 is involved, since your administrator

must grant you the permissions required to create and run flows. This is a separate concern from your Salesforce login — IAM controls "can I touch AppFlow at all," while the Salesforce OAuth connection controls "can AppFlow touch my Salesforce data."

The mental model to hold onto for your team explanation

```
Salesforce (your CRM, source of truth)
  |
  | Connection (one-time OAuth login, reused everywhere)
  v
Amazon AppFlow ← Flow = source + destination + mapping + trigger
  |
  | Destination
  v
S3 / Redshift / Amazon Connect Customer Profiles / etc.
```

That's the whole concept: **connectors** are the adapters, a **connection** is the authenticated pipe to one account, and a **flow** is a specific, configured job that moves and maps data through that pipe on some trigger.